

using the customer's credentials and stores the same on the cloud. The customer can access the *avatar* from any device (tablet, phone or PC) using a profile ID and password, and shop for any part of the wardrobe and get personalised styling suggestions.

iLUK provides users with personal styling along with the convenience of shopping anytime from anywhere. *Avatars* are accessible for a year, unless there are significant changes in measurements. Since they can try apparels on *avatars* before buying, users do not have to go through the trouble of exchanging the clothes.

Computer vision is used to capture and create 3D avatars, and digitise the garments in 3D. And AI is used to analyse body type and suggest appropriate clothing options, based on body type and fashion sensibilities.

Since the pods are available on lease, there is no capex—only monthly rental has to be paid. Other than the rental charge, there are some usage-related charges.

iLUK pods offer functional benefits to businesses, making it easy for them to engage with their customers.

—Deepshikha Shukla, technology journalist, EFY

India's First Cashier-Less Store Comes Up In Kerala

India's first cashier-less store, Watasale recently opened up in Kochi, Kerala. At the store, users need to follow three simple steps: download its app, open it and scan the unique QR code while entering the store. That's it! They can then pick the items they want and pay for these via different online payment options.

Subhash S., chief executive officer and chief technology officer, Watasale, was inspired to attempt this because of the media attention received by Amazon Go. He says, "We started working on the technology in 2015.



Watasale is India's first cashier-less store

And when Amazon announced that it will bring something similar, we became confident about what we were doing."

The system combines technologies such as QR codes, computer vision, sensor fusion, deep learning and AI. The QR code, which appears on the user's phone screen after logging in to Watasale app, acts as unique identification for him or her. After scanning it at the entrance, the system registers that the user has entered the store.

Cameras provide vision to the system, but these are fixed so that customer privacy is not compromised (no facial recognition).

Sensor fusion combines sensors like proximity, motion and weight. As soon as an item is picked up, multiple sensors complement computer vision. This gives credibility to the results. Sensors smoothly integrate into the store. These generate data for the AI-powered system to make decisions.

AI and deep learning analyse the interaction between the customer and the store.

The mobile app and its wallet system play a major role as the user exits the store. It allows him or her to pay using various wallets or credit/debit cards. Subhash adds that, since Watasale wanted customers to experience a cashier-less store, they did not

set up the prepaid bar initially. But he hints at its inclusion in the future.

The idea of an autonomous store working on AI for hassle-free shopping came up in 2015. Till 2016, in-house beta testing of the system was initiated, including hundreds of test cases. Required improvements were then followed up and different scenarios were fed by real user engagements. One interesting feature of the system is that, even if a user picks up an item and later decides not to buy it, he or she can put the item back on the shelf, without being charged for it.

After the tests and trials, an autonomous store for public preview was set up at a technology event in Kochi in 2018. And then on August 31, 2018, Watasale launched its first store in Kochi.

One major challenge faced by Watasale team was identifying the customer and purchased item(s). It did not want to implement RFID tagging, as it hinders the basic concept of pick-and-go.

Scalability of daily grocery items was also a roadblock, as these items come in all shapes and sizes. Also, depending on solely singular technologies like camera verification, sensors or tagging reduces the success percentage to about 60 per cent. **EFY**

—Nimeesh Kumar, EFY